

Since you're moving from a visual project (LED Display) to a data-driven one (Weather Logger), the best way to "create" this is to design a system that bridges the two. You can log environmental data while also displaying live updates on your LED matrix. Here is a blueprint for a complete **Intermediate Weather Logger Station**.

Project Concept: The Hybrid Weather Hub

This project will read temperature and humidity from a sensor, save it to an SD card with a timestamp, and scroll the current readings across your 8x8 LED matrix.

1. Hardware Schematic

To keep things "Intermediate," we will use an Arduino with an SD Card module and a DHT22 sensor.

Component	Function	Connection Type
DHT22	Measures Temp/Humidity	Digital Pin
SD Card Module	Stores long-term data (.csv)	SPI (MISO, MOSI, SCK, CS)
MAX7219 Matrix	Displays live readings	SPI (Shared SCK/MOSI)
RTC (Optional)	Adds real-time timestamps	I2C (SDA, SCL)

2. The Workflow (6-Hour Plan)

- **Hour 1-2 (Wiring):** Connect all three modules. Since the SD Card and LED Matrix both use SPI, they will share the same Data and Clock pins but require separate **CS (Chip Select)** pins.
- **Hour 3-4 (Code Integration):** Write the logic to read the sensor and "append" that data to a text file on the SD card.
- **Hour 5-6 (Display Logic):** Convert the numerical sensor data into a string (e.g., "Temp: 24C") and send it to the LED matrix using a scrolling function.

3. Sample Implementation Code

You will need to install the DHT, SD, and LedControl libraries.

```
#include <SPI.h>
#include <SD.h>
#include <DHT.h>
```

```
#define DHTPIN 2
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
```

```
File myFile;
```

```
void setup() {
  Serial.begin(9600);
  dht.begin();
```

```
  if (!SD.begin(4)) { // CS pin for SD card
```

```

    Serial.println("SD initialization failed!");
    return;
}
}

void loop() {
    float h = dht.readHumidity();
    float t = dht.readTemperature();

    // Log to SD Card
    myFile = SD.open("weather.csv", FILE_WRITE);
    if (myFile) {
        myFile.print("Temp: ");
        myFile.print(t);
        myFile.print(", Humidity: ");
        myFile.println(h);
        myFile.close();
    }

    // Logic to send 't' and 'h' to the LED Matrix would go here
    delay(2000); // Log every 2 seconds
}

```

Pro Tip: Data Visualization

Once you've logged data for a few hours, take the SD card out and plug it into your computer. Open the weather.csv file in Excel. You can highlight the columns and create a **Line Graph** to see how the temperature shifted in your room over time. This satisfies the "Climate Research" aspect of the learning outcome!

Would you like me to refine the code to include the scrolling text logic for the LED matrix?